



Given: Due:	Further Maths (Core Pure) HW 01	Name:			
Chapter 1 & 2: Complex Numbers & the Argand Diagram					
You MUST refer to class notes and complete personal study before attempting this homework. Attempt EVERY part of EVERY question. Check your work for errors or omissions before handing in.					
91% (A*)	76% (A)	66% (B)	56% (C)	50% (D)	40% (E)
<i>Useful resources:</i> Your notes from recent lessons & your text book <u>Although most Complex arithmetic can be done on your calculator, show all steps to reach the answer & use calculator for checking results only.</u>					
Questions - do these on lined paper					
1) A complex number z is given by $z = a + 2i$ where a is a non-zero real number. (a) Find $z^2 + 2z$ in the form $x + iy$ where x and y are real expressions in terms of a . (4) Given that $z^2 + 2z$ is real, (b) find the value of a . (1) Using this value for a , (d) Show the points P , Q and R , representing the complex numbers z , z^2 and $z^2 + 2z$ respectively, on a single Argand diagram with origin O . (3) (e) Describe fully the geometrical relationship between the line segments OP and QR . (2)					
2) The complex numbers z and w are given by $z = 8 + 3i, w = -2i$ Express in the form $a + bi$, where a and b are real constants, (a) $z - w$, (1) (b) zw . (2)					
3) $f(x) = (4x^2 + 9)(x^2 - 2x + 5)$ (a) Find the four roots of $f(x) = 0$ (4) (b) Show the four roots of $f(x) = 0$ on a single Argand diagram. (2)					



4)

$$z = 2 - i\sqrt{3}$$

(a) Show z on an Argand diagram and find the angle the line Oz makes with the positive Real axis.

(3)

Use algebra to express

(b) $z + z^2$ in the form $a + bi\sqrt{3}$, where a and b are integers,

(3)

5)

$$f(x) = (x^2 + 4)(x^2 + 8x + 25)$$

(a) Find the four roots of $f(x) = 0$.

(5)

(b) Find the sum of these four roots.

(2)

END OF HOMEWORK**Total = 32 marks**

Hand your work in with this sheet. Failure to meet your target grade will result in intervention support.

File this question paper in your file, ready for revision and file inspection.